Important Formulae & Definitions · 重要公式与定义

Topics 12, 13 & 17 (A2)

Topic 12 — Circular Motion（圆周运动）

Radian 弧度 (rad) — the angle subtended at the centre of a circle by an arc of length equal to the radius. 360° = 2π rad, 1 rad ≈ 57.3°.

Angular displacement 角位移 θ — the angle swept out by an object with respect to the centre of rotation (rad).

Angular velocity 角速度 ω — the rate of change of angular displacement: $\omega = \Delta\theta/t = 2\pi/T$ (rad s^{-1}).

Linear velocity 线速度 v — the instantaneous (tangential) speed at a point on the path: $v = 2\pi r/T = r\theta/t = r\omega$. Its direction is tangential to the circle.

Uniform circular motion 匀速圆周运动 — constant speed but changing velocity (direction changes), so there is always an acceleration, a force and a change in momentum.

Centripetal acceleration 向心加速度 — $a = r\omega^2 = v^2/r$, directed towards the centre of the circle (m s^{-2}).

Centripetal force 向心力 — the resultant of all the real forces directed towards the centre: $F = mr\omega^2 = mv^2/r$ (N). It is at right angles to the velocity, so it changes direction but not speed.

> 点评 · Insight
> 重点：向心力不是一个具体性质的力（如重力、摩擦力），而是各具体力在指向圆心方向的合效果。解题关键是先分析向心力的来源——由哪些指向圆心的力（或分量）提供。

Topic 13 — Gravitational Fields（引力场）

Newton's law of gravitation 万有引力定律 — the force between two point masses is proportional to the product of their masses and inversely proportional to the square of their separation: $F = GMm/r^2$.

Universal gravitational constant — $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

Uniform sphere — the whole mass may be taken as a point mass at its centre (r measured from the centre).

Gravitational field 引力场 — a region of space where a mass experiences a force; field lines show the direction of the force (the closer the lines, the stronger the field).

Field strength 场强 g — the force per unit mass: $g = F/m = GM/r^2$ (N kg^{-1}).

Near the Earth's surface — the field is (approximately) uniform and g ≈ 9.81 N kg^{-1} (height ≪ radius).

Effect of Earth's rotation — at the equator $mg = GMm/R^2 - mR\omega^2$ (the rotation term is tiny); at the poles $mg = GMm/R^2$.

Gravitational potential 引力势 φ — the work done per unit mass in bringing a mass from infinity to the point: $\varphi = -GM/r$ (J kg^{-1}). All values are negative (zero at infinity).

Gravitational potential energy 引力势能 — GPE = mφ = −GMm/r (J). For a large distance use ΔGPE = mφ(final) − mφ(initial); near the surface (g constant) ΔGPE = mgh.

Field–potential relation — $g = -d\varphi/dr$ = −(gradient of the φ–r graph); the minus sign shows φ decreases in the direction of g.

Circular orbit 圆轨道 — the centripetal force is provided by the gravitational force: $GMm/r^2 = mv^2/r \rightarrow v = \sqrt{(GM/r)}$.

First cosmic speed 第一宇宙速度 — the minimum orbital speed at the Earth's surface (r = R), v ≈ 7.9 km s^{-1}.

Kepler's third law — r^3/T^2 = constant (∝ the central mass M).

Total energy of a satellite — E = GPE + KE = $-GMm/r + \frac{1}{2}mv^2$ (constant for a given orbit).

Escape speed 第二宇宙速度 — $v_E = \sqrt{(2GM/r)}$: the minimum speed to escape to infinity (KE + GPE = 0).

Geostationary satellite 同步卫星 — equatorial orbit, moves west → east with period T = 24 h, so it stays over the same point of the Earth.

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天体圆轨道问题的统一思路是 “the centripetal force is provided by the gravitational force”：列 $GMm/r^2 = mr\omega^2 = mv^2/r$ 求解。$r^3/T^2 \propto M$ 说明轨道中心天体质量正比于 r^3/T^2。

Topic 17 — Oscillations（振动）

Period 周期 T — the time for one complete oscillation: $T = 2\pi/\omega$ (s).

Frequency 频率 f — the number of oscillations per unit time: $f = 1/T$ (Hz).

Angular frequency 角频率 ω — $\omega = 2\pi/T = 2\pi f$ (rad s^{-1}).

Displacement 位移 x — the distance from the equilibrium position (m).

Amplitude 振幅 x_o — the maximum displacement from the equilibrium position (m).

Phase 相位 / Phase difference 相位差 — the point reached in a complete cycle; phase difference can be given as a fraction of a cycle, in degrees or radians.

Simple harmonic motion 简谐振动 — an oscillation in which the acceleration is always proportional to the displacement and directed towards the equilibrium position: $a = -\omega^2 x$ (defining equation). ω is constant and independent of amplitude.

SHM displacement & velocity — if $x = x_o \sin(\omega t)$ (starting at equilibrium) then $v = v_o \cos(\omega t)$; if $x = x_o \cos(\omega t)$ (starting at an extreme) then $v = -v_o \sin(\omega t)$; $v_o = \omega x_o$.

v in terms of x — $v = \pm\omega\sqrt{(x_o^2 - x^2)}$; $v = v_o$ at $x = 0$, $v = 0$ at $x = x_o$.

Energy in SHM — total energy = KE + PE = $\frac{1}{2}m\omega^2 x_o^2$ (constant); PE = $\frac{1}{2}m\omega^2 x^2$, KE = $\frac{1}{2}m\omega^2(x_o^2 - x^2)$. Energy swaps between PE and KE every half-cycle.

Damping 阻尼 — loss of energy to the surroundings; light / heavy / critical damping (critical returns to equilibrium without oscillating).

Free oscillation 自由振动 — oscillating at its own natural frequency (independent of amplitude).

Forced oscillation 受迫振动 — oscillation driven by an external periodic force; it oscillates at the driving frequency.

Resonance 共振 — maximum amplitude when the driving frequency equals the natural frequency.

Effect of damping on resonance — damping lowers the resonant frequency, reduces the peak amplitude and broadens the peak.

Uses of resonance — musical instruments, microwave cooking, MRI; damaging example: earthquakes (dampers reduce the amplitude).

点评 · Insight

重点是 S.H.M 的定义：用牛顿运动定律分析振动系统，得到 a 与 x 的关系满足 $a = -\omega^2 x$
即证明是简谐振动（否则不是）。周期/频率是固定常量（natural frequency），与振幅无关。